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DEPARTMENT OF INFORMATION TECHNOLOGY
SCHOOL OF COMPUTING

10211IT204: Artificial Intelligence Techniques

"Animated Maze Solver using BFS and DFS"

PRESENTED BY

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Implementation :

- ✓ This project demonstrates how maze solving algorithms work visually.
- ✓ It shows how BFS and DFS search through a maze step by step.
- ✓ The visualization helps understand graph traversal concepts.
- ✓ Python and Pygame are used to build the animated interface
- ✓ The program highlights explored nodes and the final path.

Introduction :

- ✓ Maze solving is a common problem in computer science and artificial intelligence.
- ✓ Pathfinding algorithms help determine a route from a start point to a goal point.
- ✓ BFS and DFS are two fundamental algorithms used for graph traversal.
- ✓ These algorithms explore nodes in different ways to find a path.
- ✓ This project visualizes how these algorithms work in a grid-based maze.

Problem Statement :

- ✓ Finding a path in a maze with obstacles can be challenging.
- ✓ The program must determine whether a path exists from the start node to the goal node.
- ✓
- ✓ It must explore the grid while avoiding blocked cells.
- ✓ The algorithm should display the search process visually.
- ✓ The shortest or discovered path should be highlighted for better understanding.

Objectives :

- ✓ To understand the working of BFS and DFS algorithms.
- ✓ To create a visual representation of pathfinding in a maze.
- ✓ To compare how BFS and DFS explore nodes differently.
- ✓
- ✓ To highlight the final path from the start node to the goal node.
- ✓
- ✓ To implement the algorithms using Python and Pygame.

Concepts Used :

- ✓ To understand the working of BFS and DFS algorithms.
- ✓ To create a visual representation of pathfinding in a maze.
- ✓
- ✓ To compare how BFS and DFS explore nodes differently.
- ✓ To highlight the final path from the start node to the goal node.
- ✓ To implement the algorithms using Python and Pygame.

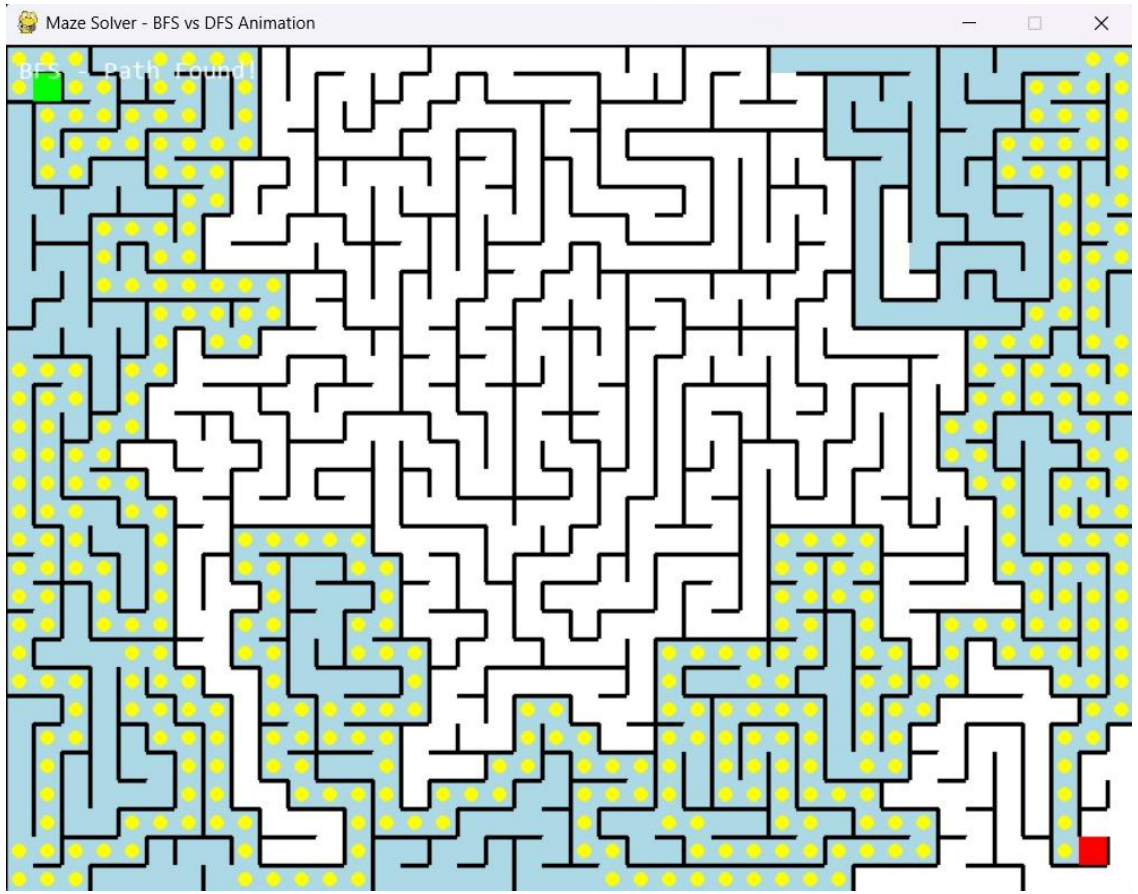
Breadth-First Search (BFS) :

- ✓ BFS explores nodes in layers starting from the start node.
- ✓ It uses a queue data structure (FIFO: First In First Out).
- ✓ All neighbors of a node are explored before moving to the next level.
- ✓
- ✓ BFS guarantees the shortest path in an unweighted graph.
- ✓ It is commonly used in navigation systems and network routing.

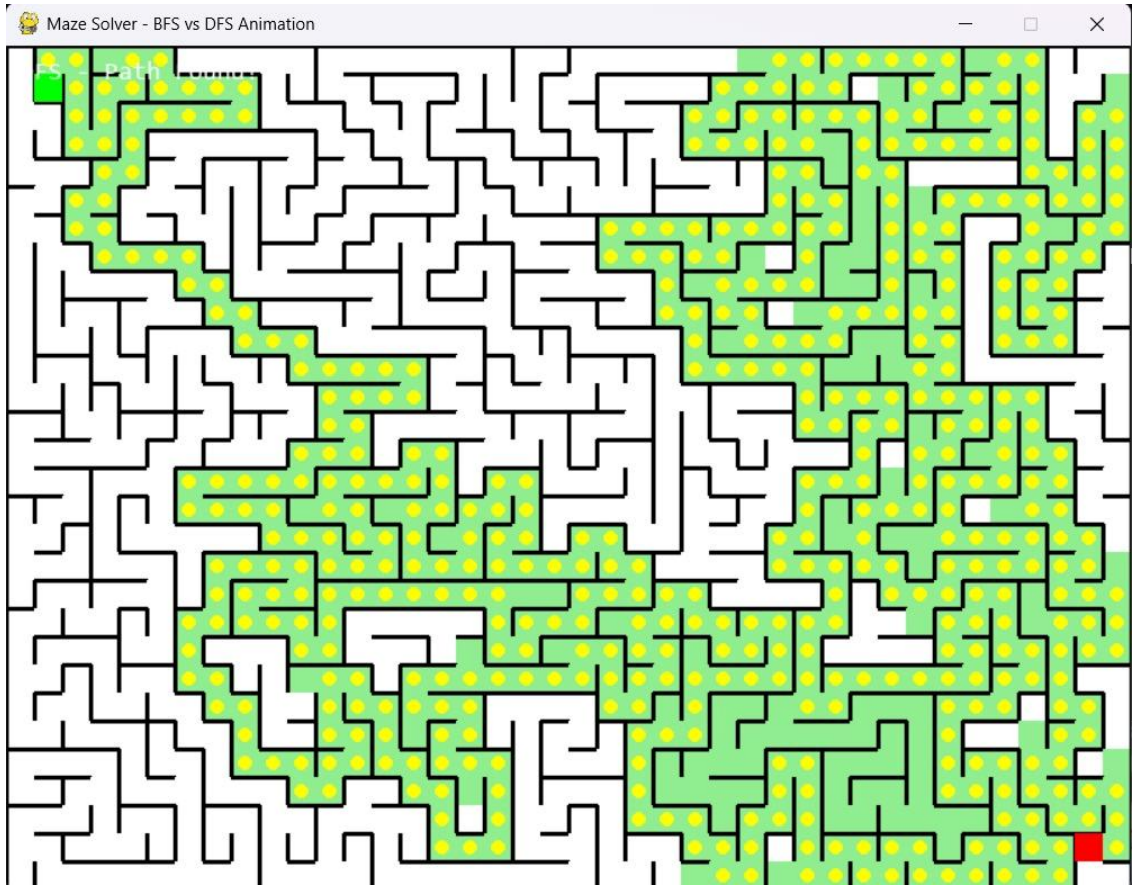
Depth-First Search (DFS) :

- ✓ DFS explores nodes by going as deep as possible along a branch.
- ✓ It uses a stack data structure (LIFO: Last In First Out).
- ✓ When a path reaches a dead end, the algorithm backtracks.
- ✓ DFS may not always produce the shortest path.
- ✓ It is useful for exploring graphs and solving puzzles.

BFS & DFS Working Diagram :



BFS



DFS

System Implementation :

- ✓ The project is implemented using Python programming language.
- ✓ Pygame is used to create the graphical interface.
- ✓ The maze is displayed as a grid with colored cells.
- ✓
- ✓ Users can place walls and run BFS or DFS algorithms.
- ✓ The program animates the search and displays the path.

Visualization Explanation :

- ✓ The maze is represented as a grid of connected paths and walls.
- ✓ The algorithm starts searching from the start node and explores the maze step by step.
- ✓ The green colored region represents the area that has been explored by the algorithm.
- ✓ The yellow dots indicate the nodes that the algorithm visits while searching for the path.
- ✓ The white sections represent unexplored areas of the maze.
- ✓ The red square represents the destination or goal node that the algorithm must reach.
- ✓ The algorithm continues exploring different paths until it finds a valid route to the goal.

Applications :

- ✓ Pathfinding algorithms are used in GPS navigation systems.
- ✓ Video games use these algorithms for character movement.
- ✓ Robotics uses pathfinding for obstacle avoidance.
- ✓
- ✓ Network routing algorithms use similar concepts.
- ✓ These algorithms are also used in artificial intelligence problems.

Conclusion :

- ✓ BFS and DFS are essential algorithms in computer science.
- ✓ They help solve pathfinding and graph traversal problems.
- ✓ The animated maze solver provides a visual understanding of these algorithms.
- ✓ BFS guarantees the shortest path while DFS explores deeply first.
- ✓ This project helps students learn algorithms in an interactive way.

THANK YOU